COMP4040 - DATA MINING AND BIG DATA ANALYTICS

REPRESENTATION LEARNING ON AMAZON REVIEWS 2023

NATURAL LANGUAGE / CLUSTERING / RETRIEVAL ANALYSIS

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AGENDA

1. Motivation
2. Dataset & Pipeline
3. Representation Methods
    a. TF-IDF Truncated SVD
    b. GLOVE - Word2Vec
    c. SBERT - BGE-Large
4. Experiments and Key Results
5. Limitations
6. Discussion and Conclusion

MOTIVATION

Computer cannot understand anything except numerical value

→ Raw text must be converted into numerical vectors

Different encoders create different vector spaces

→ How can we measure the “effectiveness” of these representation

Representation Learning: A Review and New Perspectives

Yoshua Bengio[†], Aaron Courville, and Pascal Vincent[†]
Department of computer science and operations research, U. Montreal
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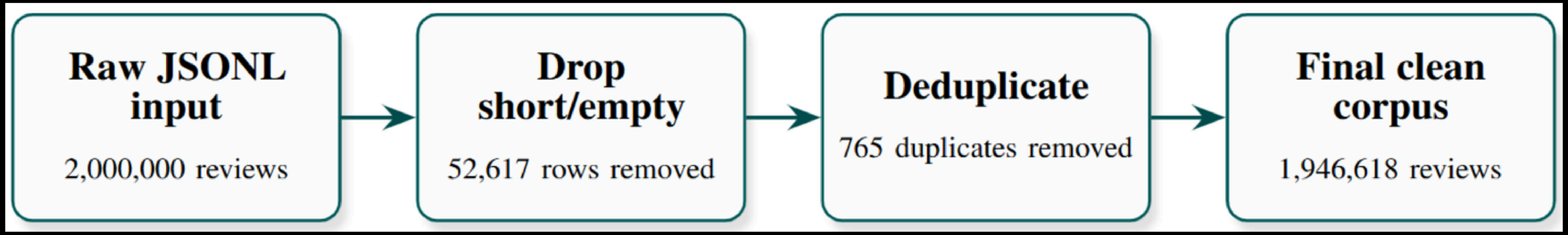
RESEARCH QUESTION

Do contextual embeddings reveal richer and more useful structure in large text corpora than sparse or static-vector representations?
And what computational cost is required to obtain these gains?

We compare classical, static, and contextual encoders on the same Amazon Reviews 2023 corpus.

DATASET OVERVIEW



Category	Raw rows	Kept rows
Books	400,000	392,155
Electronics	400,000	390,822
Kindle Store	400,000	394,313
Movies and TV	400,000	378,924
Sports and Outdoors	400,000	390,404
Total	2,000,000	1,946,618

Column	Role in this project
text	Raw review body; cleaned text is the only encoder input
title	Metadata retained for inspection; not used as an encoder input
rating	Target for the linear probe
category	External label for clustering, retrieval, and UMAP coloring
user_id	Used with product id and text for deduplication
asin, parent_asin	Product identifiers; not used as model features
verified_purchase	Metadata retained for inspection and validation

Table 1. Main dataset columns retained from Amazon Reviews 2023 and their use in the project.

Encoder input: cleaned review text only

Evaluation labels: category + star rating

OVERALL PIPELINE

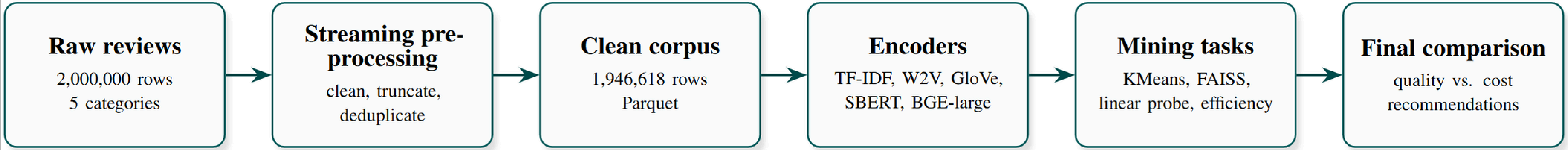


Figure 2. End-to-end experimental pipeline. The same cleaned Parquet corpus is encoded by all five representation methods, then reused for clustering, retrieval, visualization, linear probing, and efficiency analysis.

TF-IDF

SPARSE LEXICAL ENCODER

$$\text{tfidf}(t, d) = \text{tf}(t, d)\left(1 + \log \frac{1 + N}{1 + \text{df}(t)}\right)$$

- t: a term/word
- d: a document
- D: the whole dataset/corpus
- N: total number of documents
- $df(t)$: number of documents that contain term t

APPEARS OFTEN IN THIS DOCUMENT

+

DOES NOT APPEAR IN MANY OTHER DOCUMENTS

=

IMPORTANT WORD FOR THIS DOCUMENT

TRUNCATED SVD

Size of TF-IDF matrix is usually large >> We need a way to get a dense-representation instead of sparse representation >> low-rank approximation.

SVD factorizes a matrix as:

$$X = U\Sigma V^T$$

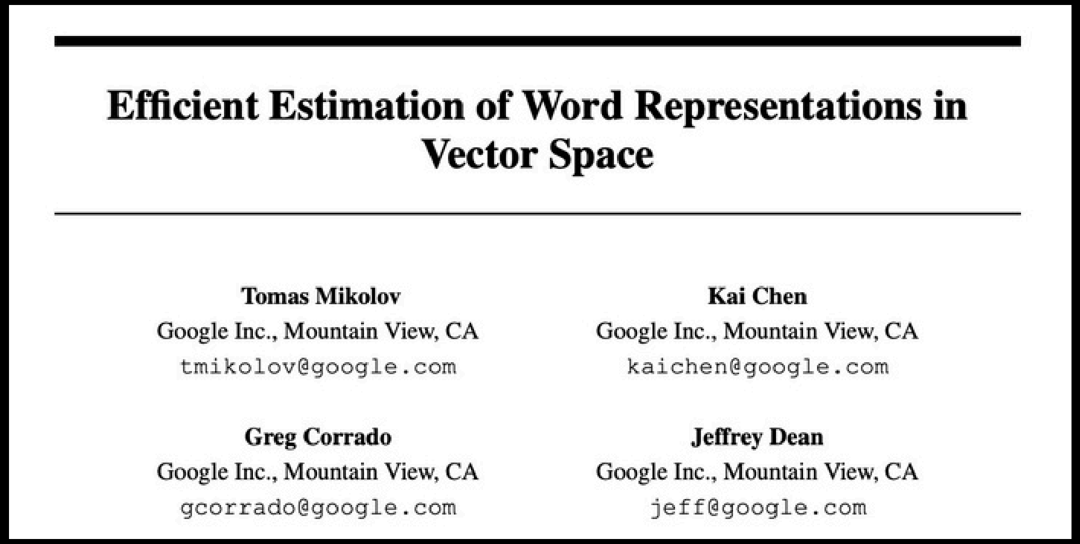
For TF-IDF:

$$X \approx U_k \Sigma_k V_k^T$$

where we only keep the top k largest singular values.

Specifically, we choose k to be 300

WORD2VEC



Efficient Estimation of Word Representations in Vector Space

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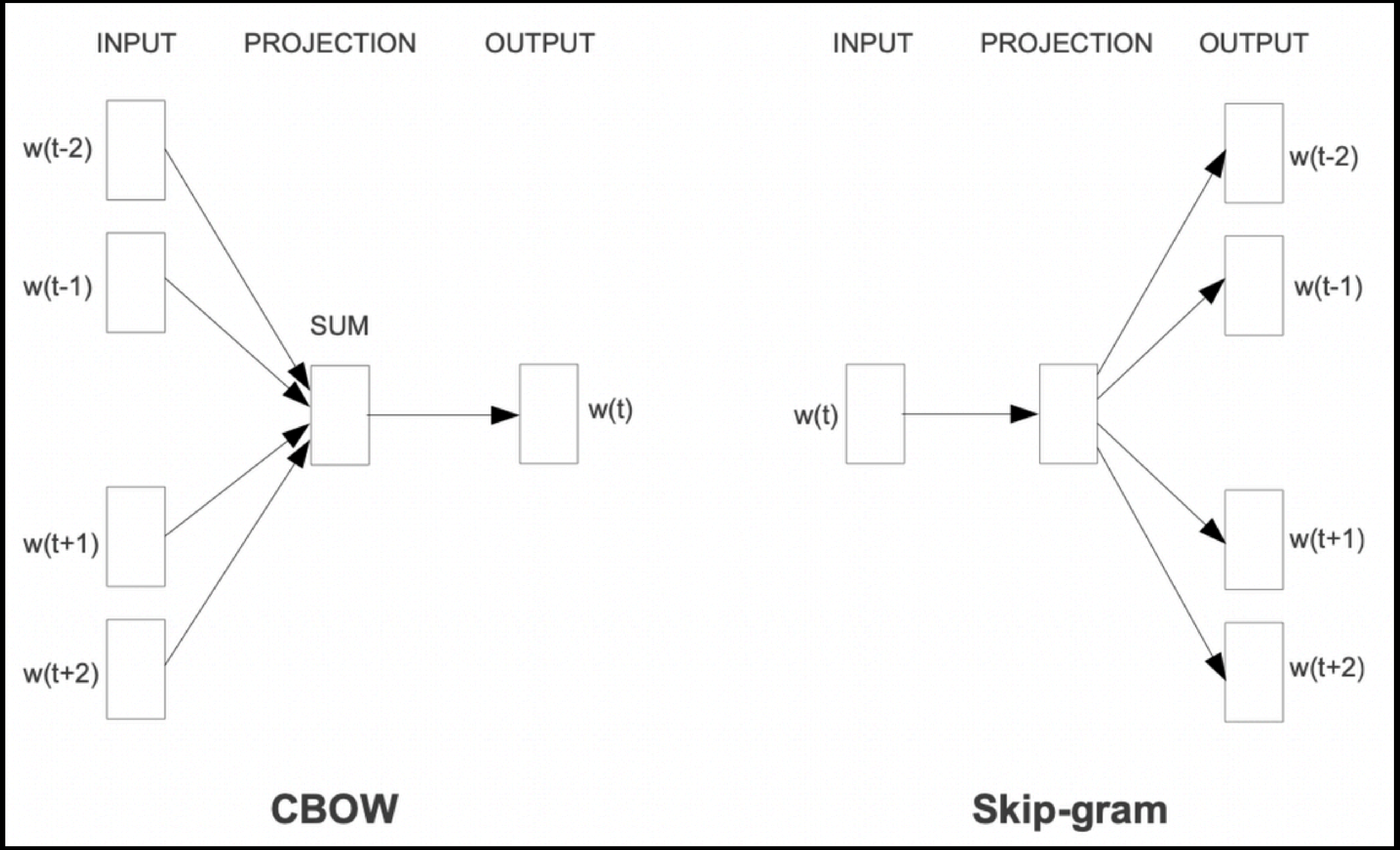
Two Architectural Variants of Word2Vec



-*"You shall know a word by the company it keeps"* -

(Firth, 1957)

Problem

- **One-hot Encoding:** A sparse, massive vector that doesn't convey information about word relationships.
- **Word2vec:** Compresses text into a dense vector space (e.g., 300 dimensions). Geometric spacing corresponds to semantic similarity.

CBOW (Continuous Bag-of-Words)

Predicts the Target word given its surrounding Context words.

Skip-gram

Predicts the surrounding Context words given a single Target word.

WORD2VEC

Large-scale Optimization

1. Softmax Probability

- Computes the conditional probability of predicting context word w_c given target word w_t via vector dot products:

$$P\left(w_c \mid w_t\right) = \frac{\exp\left(v_{w_c}'^{T} v_{w_t}\right)}{\sum_{i=1}^{V} \exp\left(v_i'^{T} v_{w_t}\right)}$$

2. Cross-entropy loss

$$J = -\log P\left(w_c \mid w_t\right)$$

The Bottleneck

Scaling for Huge Text (Google News)

Computing the Full Softmax denominator requires summing over all V words (50,000+) for every single update, causing an O(V) bottleneck.

Negative Sampling (SGNS)

Instead of evaluating all 50,000 words, the model only updates weights for 1 true context word and k randomly drawn noise samples (k in [5, 20]), reducing complexity to O(k).

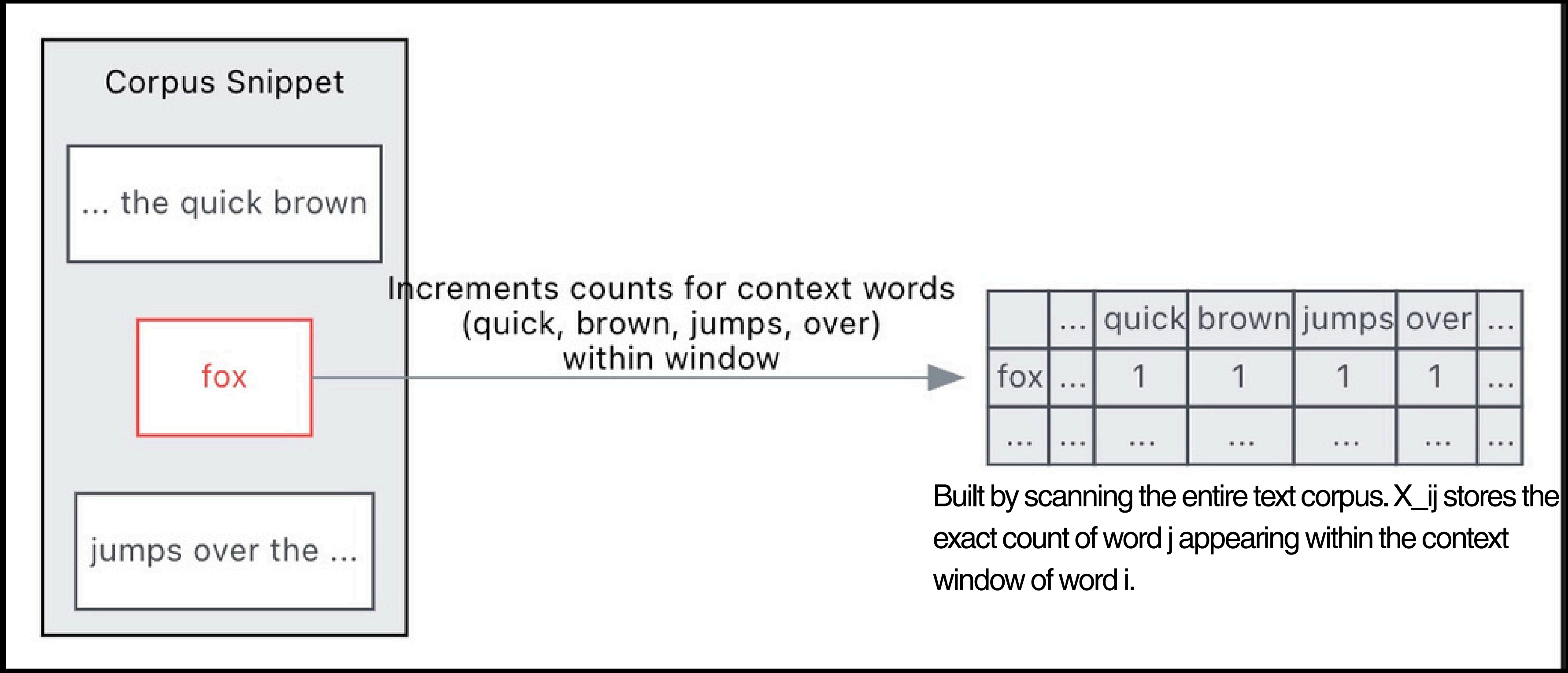
$$J_{\text{SGNS}} = -\log \sigma\left(v_{w_c}'^{T} v_{w_t}\right) - \sum_{i=1}^{k} E_{w_i \sim P_n(w)} \log \sigma\left(-v_{w_i}'^{T} v_{w_t}\right)$$

GLOVE

Global Co-occurrence & Probability Ratios

GloVe bridges the gap between Matrix Factorization (Global Statistics) and Shallow Windows (Local Prediction).

The Co-occurrence Matrix X



Built by scanning the entire text corpus. X_ij stores the exact count of word j appearing within the context window of word i.

Global Log-Bilinear Regression Loss

$$J = \sum_{i,j=1}^{V} f\left(X_{ij}\right)\left(w_i^T \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij}\right)^2$$

Core Idea: Ratios of Co-occurrence Probabilities

Probability and Ratio	$k = solid$	$k = gas$	$k = water$	$k = fashion$
$P(k\|ice)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k\|steam)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k\|ice)/P(k\|steam)$	8.9	8.5×10^{-2}	1.36	0.96

Let P_ik = X_ik / X_i be the probability of word k appearing near word i:

- For k = "solid" (related to ice, not steam): P(solid | ice) / P(solid | steam) >>> 1 (Very Large)
- For k = "gas" (related to steam, not ice): P(gas | ice) / P(gas | steam) <<< 1 (Very Small)
- For k = "water" / "fashion" (neutral / related to both / neither): P(k | ice) / P(k | steam) ≈ 1 (Near Neutral)

→ GloVe hypothesizes that these ratios encode the relationships between words, and the goal is to learn word vectors that capture these ratios.

S-BERT

CONTEXTUAL SENTENCE ENCODER

all-MiniLM-L6-v2

Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks

Nils Reimers and Iryna Gurevych
Ubiquitous Knowledge Processing Lab (UKP-TUDA)
Department of Computer Science, Technische Universität Darmstadt
www.ukp.tu-darmstadt.de

- Compact contextual encoder
- Captures review-level meaning better than averaged word vectors
- Contextual sentence encoder for semantic similarity.

- Model used: all-MiniLM-L6-v2
- 384-dimensioon sentence embedding
- Checkpoint: sentence-transformers/all-MiniLM-L6-v2.

BGE-LARGE

RETRIEVAL-ORIENTED CONTEXTUAL ENCODER

BAAI/bge-large-en-v1.5

- Maps each text into a dense vector space where semantically related reviews should be close to each other.
- Designed for high-quality semantic retrieval
- Can preserve fine-grained review meaning
- Useful when nearest-neighbor quality matters

C-Pack: Packed Resources For General Chinese Embeddings

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- Model used: BAAI/bge-large-en-v1.5.
- 384-dimensioon embedding
- Checkpoint: Hugging Face BAAI/bge-large-en-v1.5.

MAIN RESULTS

1. **Clustering : Normalized Mutual Information, Adjusted Rand Index, Silhouette Score**
2. **Semantic Retrieval : Mean Reciprocal Rank, Precision@k**
3. **Linear Probe : Accuracy, Macro-averaged F1 Score**

Embedding			Clustering			Semantic Retrieval				Linear Probe	
Encoder	Dim	Size	NMI	ARI	Sil.	MRR	P@5	P@10	P@50	Acc.	Macro-F1
TF-IDF + SVD	300	2.18GB	0.226	0.107	0.001	0.691	0.527	0.515	0.482	0.447	0.435
Word2Vec	300	2.18GB	0.014	0.010	0.006	0.722	0.580	0.569	0.546	0.432	0.420
GloVe	300	2.18GB	0.030	0.023	**0.036**	0.742	0.597	0.590	0.568	0.433	0.408
SBERT	384	2.78GB	**0.410**	**0.348**	0.016	0.819	0.718	0.712	0.697	0.456	0.443
BGE-large	1024	7.43GB	0.342	0.294	0.032	**0.828**	**0.730**	**0.725**	**0.713**	**0.590**	**0.584**

Table 3. Full-scale summary on 1,946,618 cleaned reviews. The table reports the same non-redundant quality metric set used in the static-vector ablation: clustering metrics, retrieval metrics at multiple cutoffs, and linear-probe accuracy plus macro-F1. Retrieval uses 5,000 sampled query reviews against the full FAISS index.

SBERT is best for category-aligned clustering.

BGE-large is best for semantic retrieval and rating prediction.

UMAP AND CLUSTERING

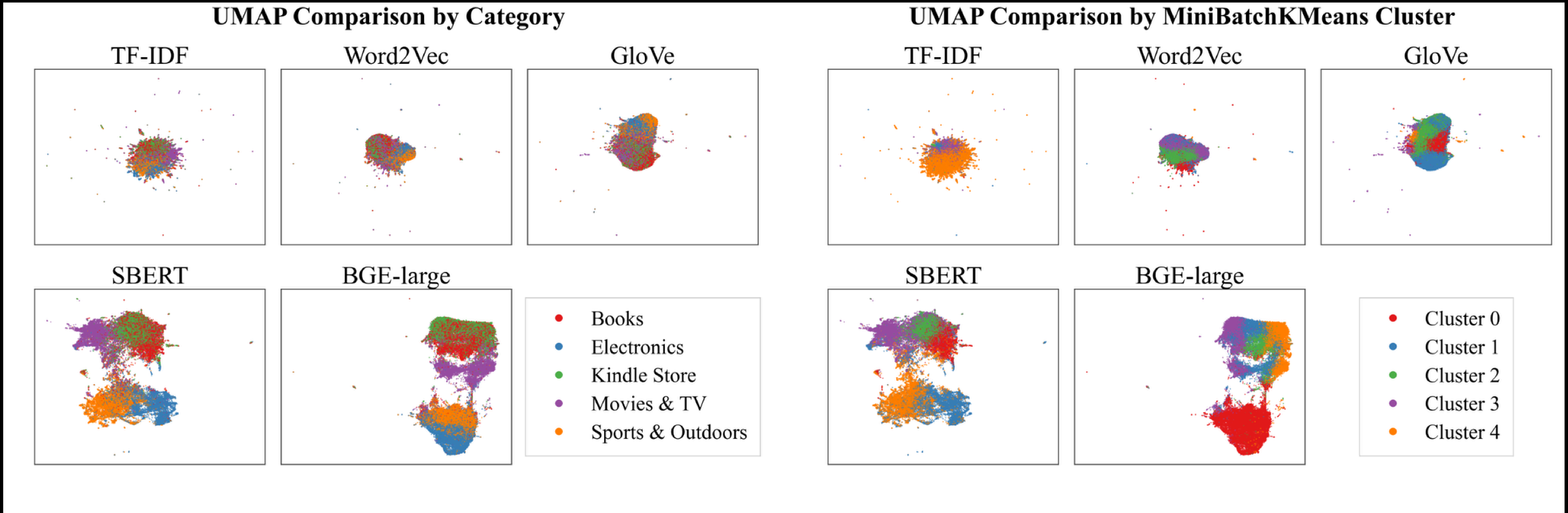

Figure 3. UMAP visualization on a 50,000-point stratified sample. Left: points colored by product category. Right: points colored by MiniBatchKMeans cluster assignment. These plots provide qualitative support for the clustering metrics.

SBERT shows the clearest category-level separation.

BGE-large creates strong local neighborhoods, useful for retrieval.

QUALITY VS. COST

- BGE-large gives the strongest retrieval and rating prediction. But it is the most expensive encoder: 1024 dimensions · 7.43GB · 65h 56m runtime
- SBERT is the better quality–cost balance: 384 dimensions · 2.78GB · 3h 01m runtime
- TF-IDF / Word2Vec / GloVe are much faster, but their representation quality is weaker.

Encoder	Device	Dim	Size (GB)	Runtime
TF-IDF	CPU	300	2.18	00:05:09
Word2Vec	CPU	300	2.18	00:02:31
GloVe	CPU	300	2.18	00:01:57
SBERT	CUDA	384	2.78	03:01:27
BGE-large	CUDA	1024	7.43	65:56:44

Table 5. Embedding efficiency summary from the full run. Runtime measures only embedding construction. Transformer encoders load pretrained weights but still compute embeddings for every review; static baselines only vectorize text or average pretrained word vectors.

ABLATION STUDY

Does training static word vectors directly on Amazon reviews improve performance?

Word2Vec improves with in-domain training.

GloVe becomes worse with in-domain training.

Static Encoder		Clustering			Semantic Retrieval				Linear Probe	
Encoder	Source	NMI	ARI	Sil.	MRR	P@5	P@10	P@50	Acc.	Macro-F1
Word2Vec	Google News	0.014	0.010	0.006	0.722	0.580	0.569	0.546	0.432	0.420
Word2Vec	Amazon reviews	**0.143**	**0.119**	0.032	**0.752**	**0.613**	**0.602**	**0.579**	**0.492**	**0.481**
GloVe	Common Crawl 840B	0.030	0.023	**0.036**	0.742	0.597	0.590	0.568	0.433	0.408
GloVe	Amazon reviews	0.007	0.004	-0.014	0.618	0.430	0.415	0.373	0.308	0.286

Table 4. Static-vector ablation comparing external pretrained vectors with vectors trained on the Amazon review corpus. The table reports the same non-redundant quality metric set as Table 3. Local runtime for the corpus-trained variants is excluded from the hardware-controlled efficiency comparison.

Domain training is not automatically helpful.

It depends on the embedding objective and the quality of co-occurrence statistics.

LIMITATIONS

- Category labels are imperfect: same-category reviews are not always semantically similar.
- Star ratings (3*) are noisy sentiment labels.
- The dataset is category-balanced, so it is not the natural Amazon distribution.
- Word2Vec/GloVe use simple averaging, losing word order and sentence meaning.
- Retrieval uses exact L2 distance only; cosine similarity or BGE prompts may change results.
- Dimensions are mismatch

CONCLUSION

- We scaled the pipeline to 1,946,618 cleaned Amazon reviews.
- Contextual embeddings capture richer structure than sparse/static methods.
- Encoder choice should depend on the task:
  - SBERT for clustering
  - BGE-large for retrieval/prediction
  - TF-IDF for lexical interpretation
- Encoder selection is a modeling decision, not just preprocessing.

THANK YOU

TF-IDF, W2V, GloVe do not have config Transformer như layers / hidden size / attention heads

Config	SBERT `all-MiniLM-L6-v2`	BGE-large `BAAI/bge-large-en-v1.5`	Meaning
Layers	6	24	BGE is 4 times deeper
Hidden size	384	1024	BGE has a wider internal representation
Attention heads	12	16	BGE uses more attention heads
FFN / intermediate size	1536	4096	BGE has a larger feed-forward layer
Output embedding dim	384	1024	BGE produces larger embeddings
Max sequence length	256	512	BGE supports longer input sequences